

# Computational Control Certificate

CTRL-01 — Vertex-copy inequality

Erdős #81, Paper II — internal validation evidence

**Reference.** Paper II, Lemma 3.1

**Date.** 2026-07-07      **Verdict.** **PASS**

## Statement audited

$\Phi_\tau(G_{v \rightarrow u}) + \Phi_\tau(G_{u \rightarrow v}) \geq 2 \Phi_\tau(G)$  for nonadjacent  $u, v$ , where  $\Phi_\tau(G) = |E(G)| - 2\tau_3^*(G)$ .

## Method

Random graphs on  $4 \leq n \leq 7$  vertices; for one random nonadjacent pair per graph, the three values of  $\tau_3^*$  are computed by an exact linear program (per-triangle covering constraints) and the inequality is checked. Seed fixed for reproducibility.

## Verbatim output

```
Random graphs n=4..7, one random nonadjacent pair each (seed 20260707)
instances tested      : 1174
inequality violations  : 0
min slack (lhs-rhs)   : -0.0000 (>=0 required; 0 => tight)
RESULT: PASS
```

## Reproducibility

The exact script `control.py` and its verbatim output `results.txt` are included in this folder. Re-running `python3 control.py` reproduces `results.txt`. This certificate reports a computational check on finite instances; it is supporting evidence, not a substitute for the analytic proof in Paper II. File integrity is recorded in `SHA256SUMS.txt` (this folder) and in the master `SHA256SUMS.txt` of the package.